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HybridCXR: Triple-Backbone CNN/Transformer Fusion with Class-Weighted Focal Loss for Robust Multi-Label Chest X-ray Diagnosis

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Graphical Abstract

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Highlights

HybridCXR: Triple-Backbone CNN/Transformer Fusion with Class-Weighted Focal Loss for Robust Multi-Label Chest X-ray Diagnosis

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- Triple-Backbone architecture (VGG16-BN/ConvNeXt/ViT) enables multi-scale feature extraction
- Novel Class-Weighted Focal Loss mitigates extreme class imbalance in chest X-ray diagnosis
- Achieves 0.870 AUC-ROC and 0.920 AUC for rare Hernia class, surpassing previous models

HybridCXR: Triple-Backbone CNN/Transformer Fusion with Class-Weighted Focal Loss for Robust Multi-Label Chest X-ray Diagnosis

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Abstract

Interpreting chest X-ray (CXR) images remains one of the most challenging tasks in medical diagnostics. The coexistence of multiple diseases in a single image (multi-label classification) and the severe class imbalance in large-scale datasets often degrade the performance of conventional deep learning models, particularly in detecting rare pathologies. To address these challenges, we propose a Triple-Backbone Hybrid Model that integrates the complementary strengths of diverse deep architectures. The proposed framework fuses the feature representations extracted from three powerful networks VGG16-BN (a classical CNN), ConvNeXt-Base (a modern CNN), and ViT-Base (Vision Transformer) through feature-level fusion. Moreover, the novel Class-Weighted Focal Loss (CWFL) function is employed to handle severe inter-class imbalance and to emphasize hard and minority examples during training. The model was evaluated on the NIH ChestX-ray14 dataset across six key thoracic diseases: Hernia, Emphysema, Edema, Pneumothorax, Effusion, and Cardiomegaly. Experimental results demonstrate that our hybrid model achieved an average AUC-ROC of 0.870, surpassing recent state-of-the-art models such as CONVFCMAE (0.852) and CheX-DS (0.838). Notably, it attained a remarkable AUC of 0.920 on the rare Hernia class, indicating its superior ability to learn discriminative and generalized features even from scarce data. These findings confirm that architectural diversity, integrated with an intelligent loss strategy, significantly enhances both robustness and generalization. The resulting framework provides a computationally efficient and highly accurate alternative to traditional ensemble methods, demonstrating strong potential

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for real-world clinical Computer-Aided Diagnosis (CAD) deployment.

Keywords: deep learning, medical image analysis, computer-aided diagnosis, multi-label classification, class imbalance, Vision Transformer

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1. Introduction

Chest X-ray (CXR) imaging is, without a doubt, one of the most widely used diagnostic tools in modern medicine. CXR remains the primary diagnostic modality for cardiorespiratory issues mainly because it is fast, low-cost, and easily accessible. However, clinically interpreting these X-rays is still a major challenge. Interpretation often involves navigating subtle visual changes and structural superposition within the chest anatomy, which makes precise judgment difficult even for expert radiologists [1]. Furthermore, the issues of inter-observer variability, a growing shortage of specialists, and the exponentially increasing volume of imaging data necessitate the development of robust, automated Computer-Aided Diagnosis (CAD) systems [2].

In the last few years, the rise of Convolutional Neural Networks (CNNs), and more recently Vision Transformers (ViTs), has profoundly impacted the field of medical computer vision [3]. Landmark achievements, such as models like CheXNet, have demonstrated that deep learning can achieve performance comparable to human radiologists in detecting critical conditions like pneumonia [4]. Despite this progress, stable and reliable performance in multi-label CXR classification is hindered by two fundamental, persistent challenges:

1. The Multi-Label and Imbalanced Data Problem:. First, the coexistence of multiple chest pathologies in a single radiograph necessitates the use of multi-label classification models that can predict several diseases concurrently. Second, large-scale datasets, such as the NIH ChestX-ray14 [5], display a 'long-tail distribution,' wherein certain diseases (e.g., Hernia) are represented by very few instances. This imbalance causes traditional models to exhibit a strong bias toward more prevalent diseases, resulting in inadequate generalization and unreliable detection of rare, yet clinically significant, conditions [5].

2. The Limitations of Single-Branch Architectures:. On one hand, CNNs are highly effective at capturing local features such as textures and edges, but struggle to model long-range spatial dependencies [6]. On the other hand, Vision Transformers (ViTs) excel at capturing these global relationships, but they lack the

strong inherent inductive biases of CNNs (e.g., translation invariance). This deficiency often necessitates larger datasets for effective learning, a common constraint in the medical domain [7].

Given these shortcomings, recent research has pivoted to hybrid models that aim to synergistically combine the respective strengths of different architectures. This approach, often termed ”architectural diversity,” posits that no single model can perfectly capture the entire spectrum of subtle disease manifestations [8]. However, even the latest hybrid models including CTransCNN, CheX-DS, and CONVFCMAE—have yet to achieve optimal balance. They struggle to maintain high performance on both prevalent classes and stable detection on rare ones. For instance, the CONVFCMAE model reported an average AUC of 0.852 [9], and CheX-DS achieved 0.838 [10]. Consequently, pushing beyond the 0.85 AUC threshold remains a significant and persistent challenge in this field.

In this study, we introduce a novel Triple-Backbone Hybrid Model termed HybridCXR. This model utilizes an ”early fusion” framework to integrate a classical CNN (VGG16-BN), a modern CNN (ConvNeXt-Base), and a Vision Transformer (ViT-Base). To counteract the extreme class imbalance, we employ the novel Class-Weighted Focal Loss (CWFL). This unified setup is specifically engineered to simultaneously model local, hierarchical, and global features.

The remainder of this paper is structured as follows: Section 2 provides a comprehensive review of previous studies and models for multi-label CXR classification, focusing on their strategies for managing data imbalance. Section 3 details our proposed methodology, including the three-branch hybrid architecture and the custom Class-Weighted Focal Loss function. Section 4 covers the dataset specifications, pre-processing steps, training setup, and evaluation metrics used. Section 5 presents the experimental results and comparative analysis against state-of-the-art models, and Section 6 concludes the paper and offers suggestions for future research.

2. Related Work

Research focused on the automated diagnosis of thoracic diseases from CXR images has experienced significant evolution over recent years[11, 12, 13, 14, 15]. The primary and persistent challenges within this domain remain the effective handling of multi-label classification and the mitigation of the severe long-tail distribution characteristic of large-scale datasets such as NIH ChestX-ray14. Despite substantial advancements in deep learning, identifying a single, universal architecture capable of simultaneously and efficiently modeling both local and global

features across the entire spectrum of pathologies remains an open problem.

2.1. CNN-Based Models: From Classic to Modern Architectures

Convolutional Neural Networks (CNNs) initiated the major shift toward automated CXR analysis. These networks are inherently proficient at extracting local, low-level features, such as texture, edge information, and image contrast. As a seminal and highly influential work, CheXNet [4, 14, 16] was introduced using the DenseNet-121 architecture, claiming to achieve radiologist-level performance in pneumonia detection, thereby instigating the new wave of research in this field. It is crucial to note, however, that subsequent replication studies often indicated that the reported performance metrics were optimistic, with actual accuracy being lower.

Subsequent developments in CXR analysis have been driven in part by modernised convolutional architectures that improve representational capacity while retaining the inductive biases of CNNs. CONVFCMAE [9], built on a ConvNeXtV2 backbone, reported a mean AUC of 0.852. Although this performance indicates strong discrimination for relatively prevalent conditions such as Emphysema and Cardiomegaly, sensitivity to infrequent pathologies remains more limited, reflecting the continuing impact of long-tailed label distributions. More broadly, despite their effectiveness for capturing local texture and edge-based cues, conventional CNNs have limited ability to explicitly model long-range spatial interactions, which may be important for characterising diffuse or spatially distributed radiographic patterns.

2.2. Transformer Models and Early Hybrid Approaches

The emergence of Vision Transformers (ViTs) provided a novel mechanism to effectively model long-range spatial dependencies in CXR images[11, 12, 17, 18, 19, 20, 21]. Models like SwinCheX [11] utilized self-attention to capture the overall image structure; however, their performance often plateaued with an average AUC of approximately 0.81. This limitation is generally attributed to their inherent lack of inductive biases compared to CNNs, often demanding substantially larger training data, which is frequently unavailable in the medical domain. Early attempts at hybrid structures, such as CTransCNN [12], which combined a CNN and a Transformer in a parallel structure, demonstrated that a simplistic architectural combination is insufficient, achieving only an AUC of 0.768 and highlighting the need for a more purposeful and integrated design.

2.3. Ensemble Methods and Managing Imbalance

Recognizing the difficulty in achieving robustness with single or simple hybrid models, other studies focused on boosting accuracy through ensemble methods and specialized loss functions. For instance, the SynthEnsemble model [13] combined outputs from CNN, ViT, and hybrid models to achieve an average AUC of 0.854. Similarly, the CheX-DS model [10] reported an AUC of 0.838 by integrating a DenseNet and a Swin Transformer, complemented by a Weighted BCE + Asymmetric Loss function.

Despite the promising accuracy figures of these approaches, they suffer from significant drawbacks: they incur extremely high computational costs for both training and inference due to running multiple networks simultaneously, and their performance against truly rare classes remains weak. Specifically, CheX-DS could only achieve an AUC of 0.789 for the critical Hernia class.

In summary, the field of multi-label CXR classification is currently constrained by three fundamental limitations[22, 23]. These include the reliance on architectures that fail to extract simultaneous local and global features; the use of loss functions that primarily rely on static, fixed weighting and thus struggle to adapt to sample difficulty and rarity; and the adoption of computationally complex ensemble models whose prolonged inference times severely restrict their practicality in real-world clinical settings.

In direct response to these specific shortcomings, this paper introduces the novel "Triple-Backbone Hybrid" model, HybridCXR. Our approach is engineered to integrate the complementary advantages of classic and modern CNNs with the powerful global modeling capacity of the Transformer architecture. Simultaneously, we employ the novel Class-Weighted Focal Loss (CWFL) to adaptively manage data imbalance and sample hardness. By addressing the lack of architectural diversity and the low sensitivity to rare classes, our proposed solution offers an efficient and highly interpretable model for automated chest disease diagnosis.

3. Methodology

This section provides a detailed technical exposition of the methodology utilized in this study. Initially, the specifics of the dataset employed for the research and the rationale for its strict partitioning into training, validation, and testing sets are introduced. This is followed by a description of the necessary pre-processing steps, including image normalization, resizing, and the applied data augmentation techniques. Next, the architecture of our novel Triple-Backbone Hybrid Model is

comprehensively detailed, along with an explanation of the feature fusion mechanism engineered to integrate feature representations from the diverse backbones. Subsequently, the specialized Class-Weighted Focal Loss (CWFL) function and the corresponding training configuration are laid out. Finally, the section concludes with a precise definition of the evaluation metrics and the process employed to assess the model's performance. The overall objective is to ensure a comprehensive and reproducible account of the design, training, and evaluation procedures for the HybridCXR model.

3.1. Dataset

The NIH ChestX-ray14 dataset was utilized for this study. It is widely recognized as one of the most comprehensive public resources for chest X-ray (CXR) image analysis. This dataset comprises 112,120 CXR images collected from 30,808 unique patients, annotated with 14 distinct disease labels [5]. The images were annotated by certified radiologists at the US National Institutes of Health. Crucially, a single image can possess one or more disease labels, defining this task as a multi-label classification problem.

Ethical considerations. This study involved secondary analysis of the NIH ChestX-ray14 dataset, which is a publicly available collection of de-identified radiographs. No new data were collected and no identifiable personal information was accessed. In accordance with the research ethics policy of our institution, analyses of fully anonymized, publicly available datasets are exempt from institutional review board oversight and do not require additional informed consent. The original data collection and public release were carried out by the US National Institutes of Health with appropriate ethical approvals and patient consent.

To maintain a focused clinical scope and evaluate model performance under more realistic conditions, we selected six key, clinically significant diseases from the full set of 14 available labels. These selected pathologies include: Hernia, Emphysema, Edema, Pneumothorax, Effusion, and Cardiomegaly. This selection was based on two primary criteria:

- **Comprehensive Coverage:** These pathologies collectively span a wide spectrum of both pulmonary (lung) and cardiac (heart) abnormalities.
- **Statistical Challenge:** They exhibit significant statistical issues, particularly severe class imbalance in the data distribution, which is a key focus of our proposed methodology.

The quantitative distribution of the six selected pathologies, including the number of samples per class, is detailed in Table 1, which clearly highlights the severe class imbalance inherent in the subset.

Table 1: Disease Distribution

Disease Name	**Count**
Cardiomegaly	2,772
Emphysema	2,516
Effusion	13,307
Hernia	227
Pneumothorax	5,298
Edema	2,303
Total	**26,423**

Although several other thoracic conditions (e.g., pneumonia) are clinically consequential, the present study prioritised a methodological objective, namely, stressing the learning problem induced by the long-tailed label distribution. Accordingly, the label subset was selected to maximise inter-class imbalance, retaining both Hernia (the rarest category; N=227) and Effusion (the most prevalent; N=13,307). This extreme frequency disparity provides a stringent test bed for the proposed Class-Weighted Focal Loss (CWFL)[24, 25], allowing a more direct assessment of its capacity to mitigate majority-class dominance and reduce overfitting to frequent labels. By contrast, including comparatively more prevalent pathologies such as pneumonia would partially attenuate the imbalance and thereby weaken the intended evaluation of CWFL under severe long-tail conditions.

As detailed in Table 1 and visualized in Figure 1, the dataset exhibits a severe degree of class imbalance. The extreme disparity is evident in the difference between the 227 samples for the Hernia class and the over 13,300 samples for Effusion. This massive imbalance risks biasing the model's weights toward the majority classes. Consequently, a specialized strategic solution specifically the implementation of the Class-Weighted Focal Loss (CWFL) was critical to ensure robust and reliable learning of discriminative features from all pathologies.

3.2. Data Preprocessing

Data preprocessing is a fundamental step to guarantee the reliable performance of deep learning models. This requirement is particularly acute for medical imag-

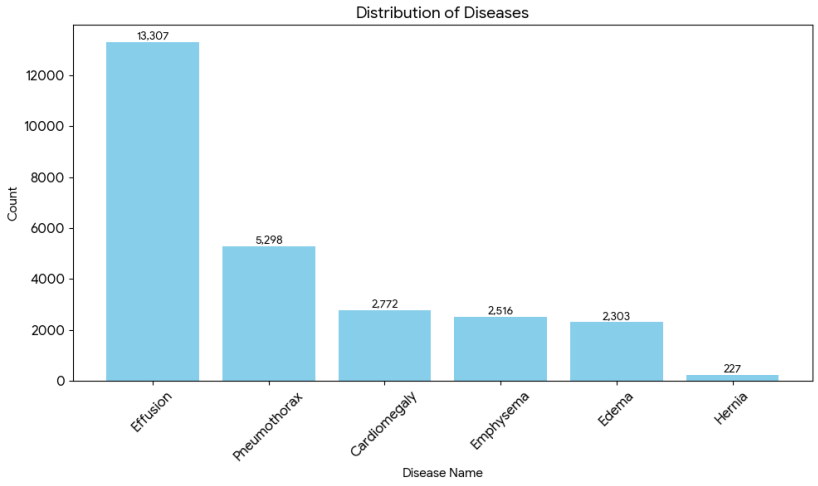


Figure 1: Disease distribution visualization showing the frequency of six thoracic conditions in the NIH ChestX-ray14 dataset (N=26,423). The bar heights illustrate the severe class imbalance between Effusion (n=13,307) and Hernia (n=227).

ing data, which frequently encounters issues such as intrinsic noise, class imbalance, and inconsistent input quality. To maximize the quality of the visual input and prevent the network from acquiring spurious or irrelevant features, a series of standard, yet optimized, preprocessing procedures were implemented.

3.3. Data Split

To ensure a rigorously fair evaluation and to completely preclude any inter-patient data leakage, the dataset was partitioned strictly based on Patient ID. As depicted in Figure 2, the determined split ratio was 70% for the training set, 15% for the validation set, and 15% for the testing set.

To effectively counter the Class Imbalance problem, especially for the highly sparse classes such as 'Hernia', we systematically deployed two core strategies:

- **Data Augmentation:** Standard image augmentation techniques were utilized, including random rotations, horizontal flips, and brightness adjustments. These methods artificially expand the dataset's variance and enhance model generalization.

- **Specialized Loss Function:** We implemented our novel Class-Weighted Focal Loss (CWFL) function. This function is specifically engineered to

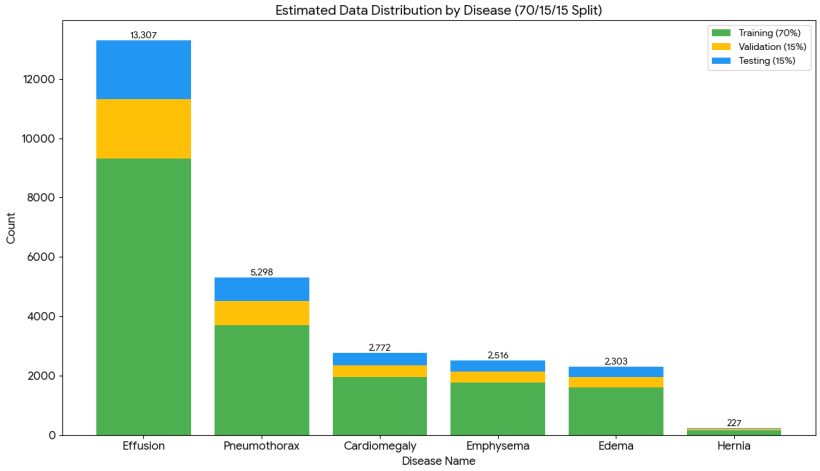


Figure 2: Data partitioning strategy visualization showing the 70-15-15 split ratio (N=26,423 images). Training set (n=18,553), validation set (n=2,709), and test set (n=5,179) were strictly separated based on Patient IDs to prevent data leakage.

dynamically re-weight samples, assigning greater importance to rare instances; the technical specifics of CWFL will be comprehensively detailed in a subsequent section.

Ultimately, our refined, partitioned dataset comprised approximately 18,553 images for training, 5,179 images for testing, and 2,709 images for validation. The exact numerical breakdown of samples for each pathology across the splits is provided in Table 2.

Table 2: Disease Distribution per Split

Disease	**Training (70%)**	**Test (15%)**	**Validation (15%)**
Hernia	158	44	25
Emphysema	1,666	577	273
Edema	1,630	460	213
Pneumothorax	3,770	971	561
Effusion	9,438	2,547	1,332
Cardiomegaly	1,891	580	305
Total	**18,553**	**5,179**	**2,709**

3.3.1. Pixel Intensity Normalization

Initially, all CXR images were converted to grayscale format, and their pixel intensity values were subsequently normalized to the standard [0, 1] range, as illustrated in Figure 3. Following this, to ensure compatibility with the specific weight initialization requirements of the pretrained models employed in this study, channel-wise normalization was executed using the mean and standard deviation parameters derived from the large-scale ImageNet dataset.

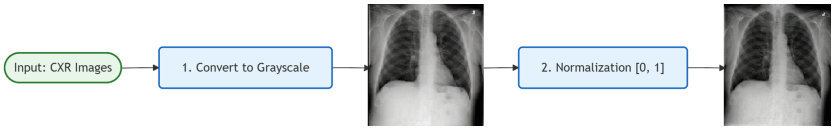


Figure 3: Pixel intensity normalization workflow for chest X-ray images. The process includes grayscale conversion and standardization to [0,1] range, followed by ImageNet-based channel-wise normalization for pretrained model compatibility.

3.3.2. Image Size Alignment

As the images within the ChestX-ray14 dataset exhibit varying dimensions, with resolutions spanning from approximately 1,000 to 3,000 pixels, all input images were uniformly resized to a fixed target dimension of 384x384 pixels. This specific dimension was selected, as illustrated in Figure 4, because it achieves an optimal balance: the resolution is sufficient to retain critical structural and textural details, while simultaneously remaining compatible with the required input specifications of all three integrated networks: VGG16-BN, ConvNeXt-Base, and ViT-Base.

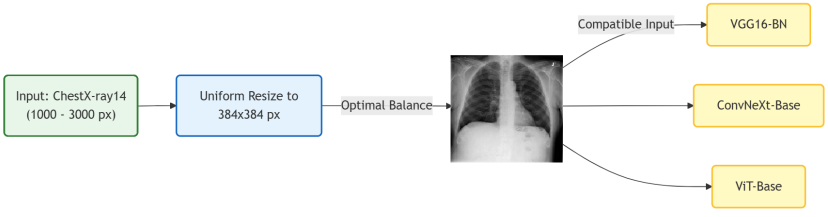


Figure 4: Image size standardization process illustrating the resizing of varied input dimensions (1,000–3,000 pixels) to a uniform 384×384 pixel resolution, optimized for the triple-backbone architecture requirements.

3.3.3. Data Augmentation

To mimic real-world clinical optimization, brightness and contrast were intentionally adjusted within a $\pm 20\%$ range at the outset of augmentation. Radiologists

routinely adjust the Window Level (brightness) and Window Width (contrast) by about $\pm 20\%$ on Picture Archiving and Communication System (PACS) setups to enhance the discernment of subtle structural details and abnormal densities. Reproducing this clinically inspired variability ensures that the model learns robust representations under realistic imaging conditions.

To artificially increase the dataset's variability and effectively mitigate the risk of overfitting, a set of random transformations was applied to the training images, each executed with a 50% probability of occurrence. These transformations encompassed:

- Random rotations (± 15 degrees)
- Random resized crop
- Horizontal flips
- Adjustments to brightness and contrast (within the 0.8 to 1.2 range)

Figure 5 demonstrates a visual example of these applied augmentations on the original images.

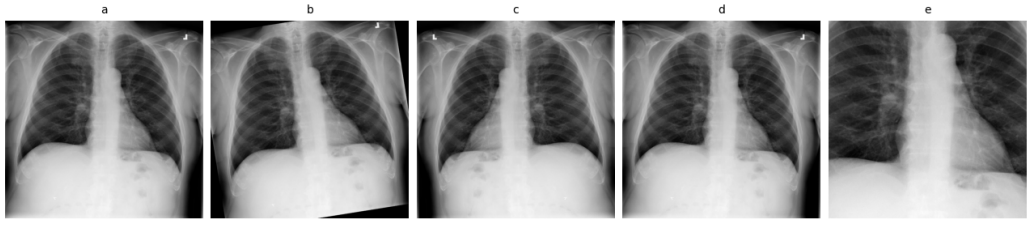


Figure 5: Data augmentation examples demonstrating the applied transformations: (a) Original image, (b) Random rotation ($\pm 15°$), (c) Horizontal flip, (d) Brightness/Contrast adjustment (0.8–1.2 range), and (e) Cropped and resized image, each applied with 50% probability.

3.4. VGG16-BN as the Local Feature Extraction Branch

The VGG16-BN network was selected as the designated branch for local feature extraction. Its deep architecture, characterized by stacked small 3×3 convolutional filters, is highly effective at capturing fine textures and intricate structural details within images. This capability is paramount for chest X-ray analysis,

particularly for the accurate demarcation of lung boundaries, detection of focal lesions, and identification of pleural effusion lines.

The integration of Batch Normalization (BN) across all convolutional layers constitutes a fundamental architectural enhancement. BN significantly stabilizes the training process, minimizes gradient instability, and contributes substantially to the model's overall performance gain. A major challenge in deep neural network training is managing unstable gradients, primarily caused by Internal Covariate Shift (ICS)—the phenomenon where the distribution of layer inputs changes during training. By incorporating BN after every convolutional layer, VGG16-BN effectively mitigates this issue.

BN operates by normalizing the activations of each mini-batch, constraining the mean and variance within a controlled range. This process directly addresses ICS, ensuring the input to subsequent layers remains statistically stable. Consequently, the network converges more rapidly, often accommodating a higher learning rate and requiring fewer epochs.

In deep sequential architectures such as VGG16, the occurrence of Vanishing or Exploding Gradients is a frequent impediment. By controlling the scale of layer activations, BN effectively prevents gradients from either dissipating or rapidly escalating. This mechanism smooths the error surface during backpropagation, leading to more stable weight updates and reduced oscillation in the loss function. Furthermore, BN enhances the model's generalization capabilities and reduces its sensitivity to weight initialization. While the original VGG could suffer from slow convergence or local minima traps due to poor initialization, the continuous normalization process in VGG16-BN ensures stable and predictable performance, regardless of the initial parameter state.

Importantly, BN also exhibits a beneficial side effect as an implicit regularizer. The statistical variations introduced by normalizing activations across different mini-batches add a form of controlled noise, which suppresses overfitting and significantly improves the model's robustness on unseen data. Figure 6 summarizes the VGG16-BN architecture.

3.5. ConvNeXt-Base as the Hierarchical Feature Extraction Branch

The ConvNeXt-Base architecture was designated as the Hierarchical Feature Extraction Branch, representing a significant modernization of Convolutional Neural Network (CNN) design principles, primarily by incorporating structural lessons derived from Vision Transformers. As summarized in Figure 7, the network utilizes greater network depth, contemporary inverted bottleneck structures, and

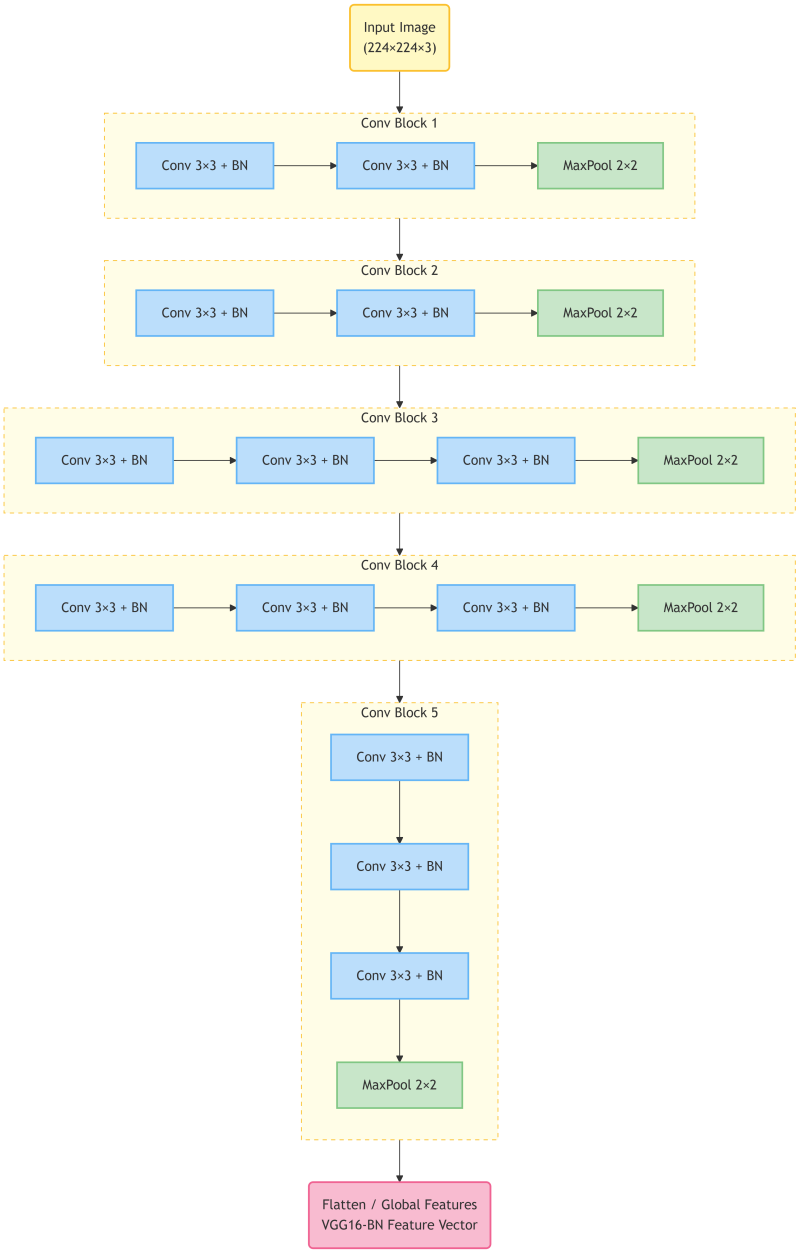


Figure 6: VGG16-BN architecture diagram showing the integration of Batch Normalization layers after each convolutional block. The network depth and filter configurations are optimized for local feature extraction in chest X-rays.

Layer Normalization. This refined configuration substantially enhances the network's capacity to model broader spatial dependencies and extract multi-scale, hierarchical context[26]. This branch therefore stands as the modern CNN component within the hybrid framework, achieving an optimal trade-off between representational capacity and computational efficiency.



Figure 7: ConvNeXt-Base architecture visualization highlighting the modern CNN design elements including inverted bottleneck structures and Layer Normalization, engineered for hierarchical feature extraction.

3.6. Vision Transformer as the Global Feature Extraction Branch

The Vision Transformer (ViT) was incorporated as the dedicated branch for global feature extraction. Unlike Convolutional Neural Networks, which primarily rely on localized pixel neighborhoods, the ViT operates on an entirely different mechanism. Its process begins by segmenting the input image into a sequence of non-overlapping patches (e.g., 16×16 pixels), which are then processed using a self-attention mechanism. This mechanism enables the model to effectively identify long-range dependencies and complex correlations between spatially disparate, or distant, areas of the image. This ability is particularly crucial for the precise diagnosis of pathologies characterized by widespread systemic patterns, such as Edema or diffuse Pneumothorax. Figure 8 illustrates the operational methodology of this ViT component.

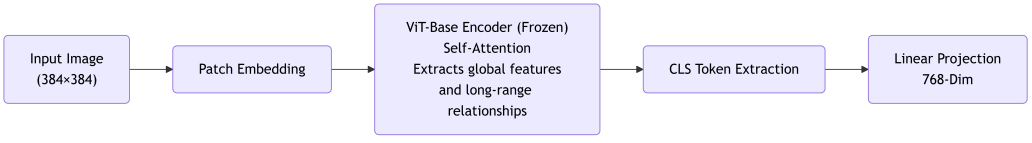


Figure 8: Vision Transformer (ViT) operational diagram showing the image patch tokenization (16×16 pixels) and self-attention mechanism for capturing global relationships in chest X-ray features.

3.7. Feature Concatenation

Following the feature extraction phase from the three separate backbone branches—VGG16-BN, ConvNeXt-Base, and ViT-Base—the final output tensor from each network was subjected to a linear projection layer. This step was necessary to map all features to a common latent dimension of 256. Once the three distinct feature vectors were standardized to this 256-dimensional space, they were subsequently concatenated (merged) along the feature axis. This operation yielded a unified 768-dimensional feature vector (256×3), which integrates a rich and complementary blend of local, hierarchical, and global representations of the chest X-ray image. This consolidated 768-dimensional vector is then, as depicted in Figure 9, passed to the model's final linear classifier layer to generate the probability predictions for each pathology.

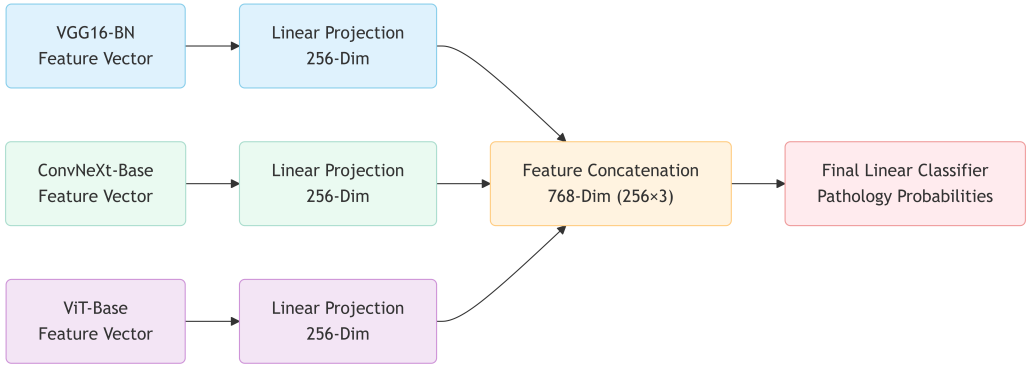


Figure 9: Feature-Level Fusion framework illustrating the concatenation of 256-dimensional feature vectors from each backbone, resulting in a unified 768-dimensional representation for final classification.

Architectural Advantage. The principal advantage of employing this Feature-Level Fusion approach, particularly when contrasted with simpler Late Fusion strategies such as output averaging, lies in its capacity to preserve the richness and complexity of the information[27, 28]. By combining the feature vectors prior to the final classification decision, the architecture facilitates more complex and non-linear interactions between the distinct local, hierarchical, and global features extracted by the three backbones. This mechanism yields a significantly more comprehensive, accurate, and robust final representation of the input image. Fundamentally, this method demonstrably improves the retention of detailed information and substan-

tially amplifies the synergistic interaction between features, leading to enhanced overall discriminative power compared to mere score aggregation.

3.8. Classification Layer

The consolidated feature vector is subsequently fed into a final Fully Connected (FC) layer. This layer is configured with six output nodes, corresponding precisely to the six target pathologies of the study. The Sigmoid function was selected as the final activation for the output layer. This choice is crucial as it enables the model to predict the probability of each disease independently, which is the requisite mechanism for operation within a multi-label classification framework. The operational flow of this final classifier is illustrated in Figure 10.

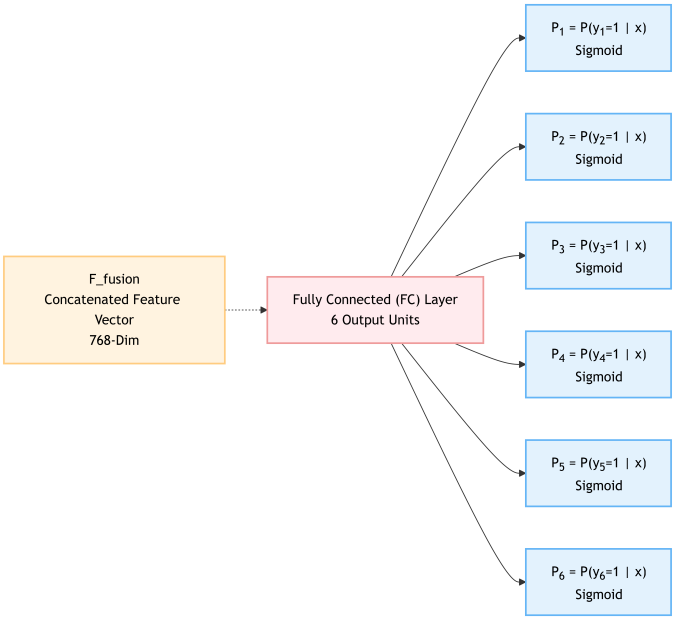


Figure 10: Classification layer diagram showing the transformation of the 768-dimensional fused feature vector into independent disease probability predictions through Sigmoid activation for multi-label classification.

The prediction for each pathology label is mathematically defined by the following expression, as shown in Equation (1):

$$P_i = \sigma(W_i \cdot F_{fusion} + b_i) \tag{1}$$

where $P(y_i = 1|x)$ is the probability of the i-th disease being present and σ is the sigmoid function.

3.9. Key Features of the Proposed Model

The proposed HybridCXR model incorporates several core architectural innovations designed to maximize diagnostic accuracy and robustness:

- **Complementary Triple-Backbone Structure:** The design synergistically integrates a classical CNN (VGG16-BN), a modern CNN (ConvNeXt-Base), and a Vision Transformer (ViT-Base) to ensure feature extraction across all scales—local, hierarchical, and global.
- **Adaptive Feature Fusion:** The framework utilizes an adaptive feature fusion mechanism, where the concatenation of projected feature vectors enables complex, learnable interactions between these diverse feature representations before the final classification.
- **End-to-End Efficiency:** Despite its three parallel branches, the model maintains high computational efficiency as it is designed for end-to-end training, eliminating the need for external or time-consuming ensembling methods.
- **Generalization Capability:** This design confers high generalization capability, demonstrated by robust performance across all pathologies, including particularly strong results on highly rare classes such as Hernia.

3.10. Ablation and Model Selection Study

Prior to finalizing the three-branch architecture, a systematic ablation study involving several distinct network combinations was executed. The primary objective was to identify the optimal configuration that yielded the best trade-off between diagnostic accuracy, generalization capacity, and computational complexity. This procedure was essential to quantify the individual contribution of each architectural type—classic CNN, Modern CNN, and Transformer—to the overall predictive performance of the final model. Table 3 presents the mean AUC-ROC results obtained from the various network combinations evaluated.

Table 3: Mean AUC-ROC from Various Network Combinations

Model Combination	**Combination Type**	**Mean AUC**	**Performance Explanation**
VGG16-BN + ResNet152V2	Two-Branch CNN	0.842	Good local feature extraction, but weak on global image relationships
ConvNeXt-Base + ViT-Base	Modern CNN + Transformer	0.857	Better performance on scattered patterns, but drops on small classes
VGG16-BN + ViT-Base	Classic CNN + Transformer	0.861	A decent balance, but training convergence was unstable
DenseNet201 + ConvNeXt-Base	Two-Branch Modern CNN	0.848	Lots of feature overlap, no useful diversity
HybridCXR	**Three-Branch**	**0.870**	**Best balance of accuracy and stability**

Model Selection Analysis. The results presented in Table 3 unequivocally demonstrate that fusing all three complementary architectures—the classic CNN, the modern CNN, and the Vision Transformer—yields a significant enhancement in both mean AUC-ROC and training stability. Two-branch combinations consistently exhibited inherent limitations, often resulting in a notable deficit in either predictive accuracy or convergence stability. Conversely, the triple-branch structure successfully established an optimal equilibrium by effectively integrating local, hierarchical, and global feature representations. Consequently, the HybridCXR three-branch hybrid architecture was selected as the optimal and final model for all subsequent experiments and evaluations conducted in this study.

3.11. Fusion Strategy Ablation Study

While concatenating the projected feature vectors is the primary fusion strategy used in the HybridCXR model, alternative feature-level aggregation methods were evaluated. We compared simple mean averaging and max pooling across the three backbone feature vectors against concatenation to quantify the impact of the fusion mechanism independent of backbone selection. Table 4 summarizes the results and shows that concatenation consistently outperformed these simpler aggregation methods, underscoring that preserving the full diversity of local, hierarchical, and global representations yields superior performance.

Table 4: Ablation of Feature Fusion Strategies

Fusion Method	**Mean AUC**	**F1-Score**	**Explanation**
Concatenation	0.870	0.811	Combines features into 768-dim representation
Feature Averaging	0.862	0.804	Smooths signals, reduces discrimination
Feature Max Pooling	0.857	0.799	Suppresses subtle patterns

3.12. The Proposed CWFL Loss Function

Addressing severe class imbalance is one of the paramount challenges in developing robust deep learning models for multi-label medical image classification, particularly with chest X-ray data[29, 30, 31, 32, 33]. Pathologies like Effusion exhibit thousands of training instances, while minority classes such as Hernia are represented by only a few hundred samples. In this scenario, standard loss functions permit the model to become heavily biased towards high-frequency classes, effectively suppressing the learning signal from critical rare instances.

To counteract this critical issue, this study proposes a novel objective function termed Class Weighted Focal Loss (CWFL). This loss function strategically integrates two highly effective strategies for imbalance control: inverse class frequency weighting and the Focal Loss mechanism, which prioritizes the learning of difficult, misclassified examples.

The proposed CWFL function is mathematically formulated as the summation over all samples (N) and classes (C), as shown in Equation (2):

$$CWFL = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} w_c \alpha_c (1 - p_{ic})^{\gamma} y_{ic} \log(p_{ic}) \tag{2}$$

In this equation, N is the total number of samples, C is the number of classes, $p_{i,j}$ is the predicted probability for sample i in class j, $y_{i,j}$ is the true label (0 or 1), and w_j is the class weight, which is calculated based on the inverse frequency of each class in Equation (3):

$$w_c = \frac{1/f_c}{\sum_{k=1}^{C} 1/f_k} \tag{3}$$

where f_c is the number of samples in class C, α_c is the focal balancing factor (set to 0.25 in this study), and γ is the focusing parameter that controls the focus

on hard-to-classify samples (which we set to 2.0).

In this loss function, the $(1 - p_{ic})^{\gamma}$ factor is key. It reduces the weight of the "easy" samples and forces the model to focus on the harder, rarer ones. At the same time, the w_c class weight boosts the contribution of those low-frequency diseases to the overall gradient. Combining these two components makes the model much more sensitive to critical cases at both the Class-level and the Instance-level.

The nomenclature of the method is derived directly from its core components: Class-Weighted refers to the incorporation of weights based on class frequency, and Focal denotes the mechanism that dynamically modulates each example's loss contribution. Although both elements are well established in the deep learning literature, this study integrates them in a single objective tailored to the extreme label imbalance of the NIH ChestX-ray14 dataset. In our experiments, this combined formulation was necessary to stabilise optimisation of the multi-branch architecture by reducing majority-class dominance and limiting the attenuation of gradients associated with rare labels (e.g., Hernia).

3.13. Training Settings

The development environment utilized a Jupyter Notebook interface, with the model implementation and training conducted using PyTorch 2.1.0. Code version control and documentation were managed via a dedicated GitHub repository. CUDA 12 was employed for essential GPU acceleration during all computational procedures.

3.13.1. Hardware and Development Environment

All model training and evaluation procedures were executed within the standardized environment detailed in Table 5.

Table 5: Hardware Configuration for Model Training

Component	**Specifications**
GPU	1 x NVIDIA RTX 4090 (24GB VRAM)
CPU	16-Core Processor
RAM	32 GB
Disk	150 GB SSD

3.13.2. Libraries Used

Table 6 lists the primary libraries utilized across the different stages of the project development and evaluation.

Table 6: Libraries Used

Category	**Libraries**
Deep Learning Framework	torch, torchvision, torch.nn, torch.optim
Advanced Models	timm, transformers (for ViT)
Data & Numeric Processing	numpy, pandas
Image Processing	PIL (Pillow)
Machine Learning & Evaluation	scikit-learn (sklearn.metrics, model_selection)
Plotting & Analysis	matplotlib, seaborn
Utility Tools	tqdm (for progress bars), os, random

3.13.3. Training Configuration

Table 7 summarizes the critical parameters and hyperparameters established for model training. It is pertinent to note that the feature extraction layers of the pretrained backbones were maintained in a frozen state. Training was exclusively applied to the newly introduced layers following the feature fusion stage (e.g., the final classifier).

3.13.4. Training and Evaluation Process

To ensure result stability and reproducibility, the model was trained three separate times, and the final reported performance represents the average across these runs. The training process was executed through a sequence of critical steps. Initially, class weights were calculated based on the label distribution within the dataset. Subsequently, the CWFL loss function was applied to compute the loss for every batch. Training convergence was managed by implementing Early Stopping, which monitored improvements to the Macro AUC score on the validation set. The final model weights corresponding to the highest performance achieved on the validation set were then saved. Finally, all training metrics, including the evolution of Loss, AUC, and F1-score across all epochs, were systematically logged and analyzed.

3.14. Evaluation Metrics

To obtain a comprehensive assessment of the proposed model's performance in this multi-label classification task, a specific set of statistical metrics was utilized, as detailed in Table 8. Each of these metrics is employed to quantify a distinct aspect of the predictive quality and discriminative ability of the model.

Table 7: Model Training Configuration

Parameter	**Value**	**Explanation**
Model Architecture	HybridCXRModel	Triple-branch structure with frozen feature extractors
Dataset	NIH ChestX-ray14	Using 6 key diseases with a patient-centric split
Image Input Size	384 x 384 pixels	To preserve structural details and match model inputs
Batch Size	128	Sized to fit the RTX 4090 GPU memory
Epochs (Max)	30	Hit its best performance at Epoch 21, triggered early stopping at Epoch 26
Patience	5 epochs	Monitored the Macro AUC on the validation set
Optimizer	Adam	Used its adaptive learning rate for faster convergence
Initial Learning Rate	1e-4	We didn't use a scheduler; we just lowered it manually if we saw fluctuations
Loss Function	Weighted Focal Loss ($\gamma = 2.0$)	To combat the severe class imbalance
Output Activation	Sigmoid	The standard choice for multi-label classification
Evaluation Metrics	Various (see Table 8)	To get a comprehensive look at the model's performance

Table 8: Evaluation Metrics

Metric	Definition	Explanation
AUC-ROC	The area under the ROC curve, calculated for each label	Measures the model's ability to discriminate between positive and negative samples for each disease
AUC-PR	The area under the Precision-Recall curve	Great for highly imbalanced data. It's sensitive to performance changes in the rare classes
Precision	Ratio of correct positive predictions to all positive predictions	Shows how accurate the model's positive predictions are
Recall	Ratio of correct positive predictions to all actual positives	A measure of the model's ability to "catch" all the real patients
F1-Score	Harmonic mean of Precision and Recall	A balanced metric that considers both Precision and Recall
Accuracy	Ratio of all correct predictions to total samples	Useful for balanced data, but can be misleading in imbalanced datasets
Macro AUC	Average AUC score across all classes	This was the main metric we used to compare different models in this study

The final output for each disease probability was derived from the Sigmoid activation function, yielding a score within the [0, 1] range. Subsequently, the Optimal F1 Threshold was calculated for each individual label using the validation dataset. This determined threshold was then applied consistently during the final evaluation of the test dataset.

4. Results and Analysis

The initial phase of the experimental evaluation involved a systematic ablation study to determine the optimal backbone combination. The results, as previously presented in Table 3, unequivocally demonstrated that fusing all three complementary architectures—the classic CNN (VGG16-BN), the modern CNN (ConvNeXt-Base), and the Vision Transformer (ViT-Base)—yielded a significant enhance-

ment in both mean AUC-ROC and training stability. Two-branch combinations consistently exhibited inherent limitations, often resulting in a notable deficit in either predictive accuracy or convergence stability. Conversely, the triple-branch structure successfully established an optimal equilibrium by effectively integrating local, hierarchical, and global feature representations. Consequently, the HybridCXR three-branch hybrid architecture was selected as the optimal and final model for all subsequent experiments and evaluations conducted in this study.

4.1. Evaluation Results

The performance of the proposed HybridCXR model was rigorously evaluated on the test set across the six target pathologies using the comprehensive set of metrics defined in Table 8. The model achieved a mean AUC-ROC of 0.870 and an F1-score of 0.811, demonstrating a robust balance between precision and recall (sensitivity) across the multi-label task. The detailed performance results for each of the six target diseases are presented in Table 9.

Table 9: HybridCXR Results for the Six Target Diseases

Disease	**AUC-ROC**	**F1-Score**	**Precision**	**Recall**
Hernia	0.920	0.848	0.861	0.835
Emphysema	0.890	0.829	0.822	0.836
Edema	0.880	0.804	0.796	0.812
Pneumothorax	0.850	0.783	0.776	0.789
Effusion	0.840	0.792	0.787	0.798
Cardiomegaly	0.840	0.810	0.818	0.802

The results clearly indicate exceptional performance on the Hernia class, which is one of the rarest pathologies in the NIH ChestX-ray14 dataset, achieving an AUC of 0.920. This result strongly validates the efficacy of the Class-Weighted Focal Loss (CWFL) function and the multi-branch structure in extracting discriminative features even from severely under-represented samples. Conversely, diseases like Effusion and Pneumothorax, which possess higher visual diversity, demonstrated slightly lower yet still strong performance, remaining within the AUC ¿ 0.84 range.

To gain a finer understanding of the model's discriminative ability and performance patterns, the ROC curve for each individual class was plotted and analyzed, as presented in Figure 11.

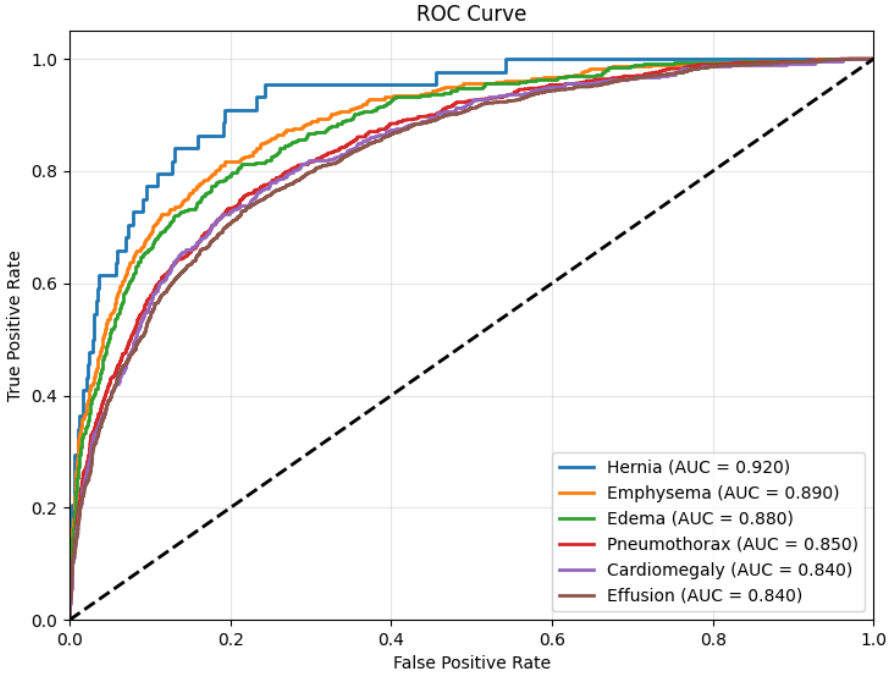


Figure 11: Receiver Operating Characteristic (ROC) curves for the six target diseases. Each curve shows the trade-off between sensitivity and specificity, with the area under curve (AUC) ranging from 0.840 (Effusion) to 0.920 (Hernia).

Furthermore, a Correlation Matrix depicting the relationships between the target labels was calculated and visualized, as shown in Figure 12.

The analysis of this matrix confirms strong epidemiological and radiological overlaps, particularly between Effusion–Edema and Cardiomegaly–Edema, which aligns with existing clinical knowledge.

4.2. Comparison with Previous Studies

To benchmark the HybridCXR model, its performance was compared against several well established and contemporary architectures, including CNN-based, Transformer-based, and Hybrid models. This comparative analysis provides a clear context for judging the relative advantages of the proposed framework in the field of multi-label chest disease classification. Table 10 summarizes the performance of the HybridCXR model alongside these benchmark studies on the NIH ChestX-ray14 dataset.

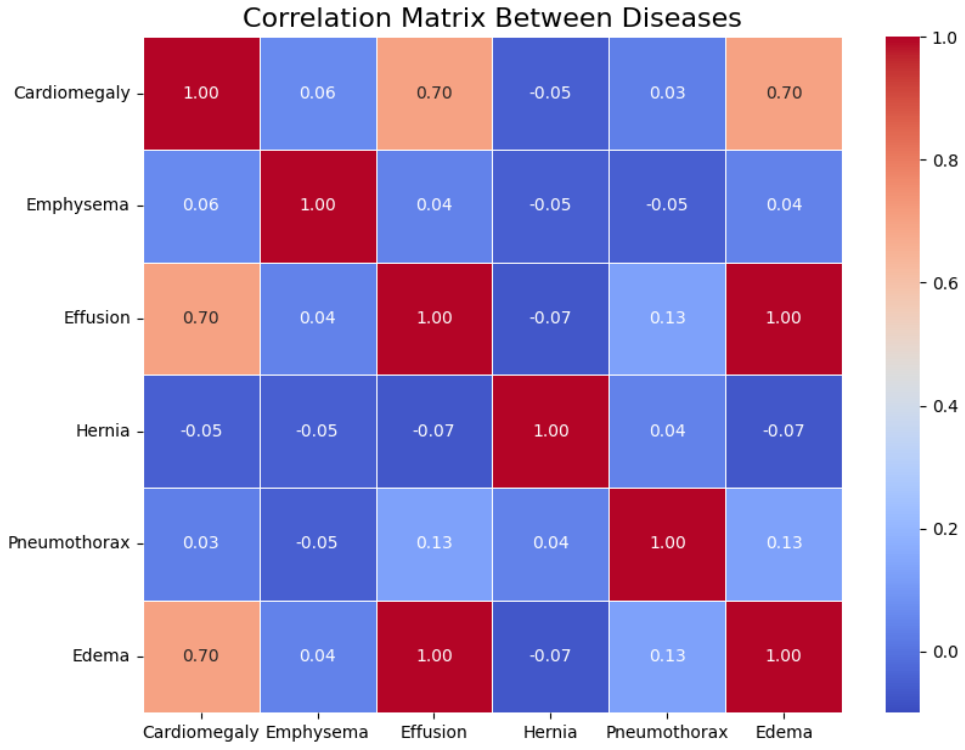


Figure 12: Disease correlation matrix computed from the NIH ChestX-ray14 dataset (n=26,423). Color intensity represents the strength of association between pathology pairs, with darker shades indicating stronger correlations.

4.3. Computational Efficiency Analysis

Beyond predictive performance, practical deployment of Computer-Aided Diagnosis (CAD) systems depends on inference speed and memory footprint. Because HybridCXR fuses three backbones into a single end-to-end network, it avoids the latency and memory overhead associated with running separate models in sequence. Table 11 compares HybridCXR with representative ensemble-based approaches in terms of inference time per image and approximate parameter count. Despite integrating three feature extractors, HybridCXR requires fewer parameters and substantially less inference time than ensembles that load and execute multiple independent networks.

4.4. Interpretability and Robustness to Radiographic Artifacts

We acknowledge the concern that high-capacity models may partially rely on radiographic artifacts (e.g., embedded text markers, support devices, or anatom-

Table 10: Comparison of HybridCXR with Previous Studies

Model	Architecture	AUC	F1	Prec.	Key Features
CheXNet	DenseNet-121	0.841	0.762	0.755	First deep network for CXR classification
CheXpert	DenseNet-121	0.854	—	—	Framework with uncertain labels
CXRs-HAM10000	CNN+Attention	0.861	—	—	Includes location information
SwinCheX	Transformer	0.810	0.748	0.755	Global relationship modeling
CTransCNN	Hybrid	0.768	0.701	0.707	CNN-Transformer combination
SynthEnsemble	Ensemble	0.854	0.782	0.775	Multi-model fusion approach
CheX-DS	Ensemble	0.838	0.777	0.774	DenseNet-Swin combination
MedViT	ViT+CNN	0.855	—	—	Lightweight hybrid design
CONVFCMAE	ConvNeXtV2	0.852	0.794	0.790	Modern CNN architecture
HybridCXR	**Triple-Backbone**	**0.870**	**0.811**	**0.806**	**Optimal feature integration**

Table 11: Inference Efficiency and Model Size Comparison

Model/Approach	Time (ms/img)	Params	Explanation
HybridCXR	**≈27**	**≈180M**	Single fused network; branches share memory and computation
CheX-DS	≈65	≈200M	Two independent networks; requires sequential inference
SynthEnsemble	≈81	≈300M	Weighted ensemble; inference time scales with models

ical shadows) rather than pathology-related image cues. Although a comprehensive post hoc interpretability analysis (e.g., Grad-CAM) is outside the scope of the present work and is therefore deferred to future research, several aspects of the experimental design reduce the risk of artifactual shortcut learning.

First, the patient-level data partitioning strategy (Section 3) provides an important control against memorisation of patient-specific or acquisition-specific patterns. By ensuring that no patient appears in more than one split, the model is prevented from exploiting stable, individual-level markers that could otherwise inflate test performance under image-level splitting.

Second, the inclusion of the ViT-Base branch may contribute to increased robustness in this context. In contrast to purely local convolutional processing, the global self-attention mechanism encourages the model to integrate information across the full lung field and mediastinal region, which may reduce the likelihood that predictions are driven predominantly by a small, localised artifact that a CNN-only backbone could overemphasise.

4.5. Comparative Discussion

The analysis of the results presented in Table 10 highlights several significant trends in the progression of deep learning techniques for CXR diagnosis. Initial CNN-based models, such as CheXNet [4], achieved satisfactory performance in identifying common diseases but faced considerable challenges in detecting rare pathologies. This limitation is largely due to their dependence on traditional loss functions, which are particularly sensitive to data frequency and diminish the gradient contribution from infrequent samples.

In response to these challenges, Transformer-based models, including SwinCheX [11], employed attention mechanisms to effectively capture global dependencies. While this approach is beneficial for addressing complex, non-local patterns, these models often encountered difficulties related to a loss of local resolution and high computational demands.

Hybrid methods, such as CTransCNN [12], sought to merge CNN and Transformer features through cross-attention mechanisms. However, this approach yielded a notably lower AUC of 0.768 compared to the 0.870 achieved by our model. We attribute this performance gap to the differences in parameter efficiency associated with the fusion strategy: while cross-attention theoretically facilitates dynamic feature querying, it also introduces additional learnable parameters that can increase model complexity and exacerbate overfitting, particularly when training data are limited, as is common in medical imaging. In contrast, our proposed direct concatenation-based fusion method maintains the distinctiveness of local and global feature spaces without the added complexity and potential instability of attention layers during optimization.

4.6. Superiority of the HybridCXR Approach

The proposed HybridCXR model achieved the best performance among all compared architectures, with a mean AUC of 0.870. This superiority stems from its dual-level innovation:

- **Triple-Backbone Feature Integration:** The use of three complementary backbones VGG16-BN (local features), ConvNeXt-Base (efficient deep representations), and ViT-Base (global relationships)—successfully mitigates the trade-off inherent in single-architecture designs. This architectural diversity is highly effective for multi-scale feature learning, particularly in challenging, low-sample classes.

- **Adaptive CWFL Loss Function:** The CWFL function strategically combines Class-weighting based on inverse frequency to amplify the importance of rare diseases, and the Focal mechanism to dynamically focus the model on hard instances. This dual-level design stabilizes the gradient and ensures an adaptive focus, overcoming the limitations faced by previous models using standard or non-adaptive loss methods.

Notably, the 0.018 AUC improvement over the modern CNN-based CONVFCMAE (0.852 AUC) underscores the necessity of architectural diversity. Whereas CONVFCMAE relies solely on a highly optimized ConvNeXtV2 backbone, HybridCXR supplements a similar modern CNN (ConvNeXt-Base) with both the fine-grained inductive biases of VGG16-BN and the global self-attention mechanisms of ViT-Base. This complementary combination captures subtle local structures and long-range dependencies that a single advanced CNN overlooks, leading to superior detection of underrepresented diseases such as Hernia.

Furthermore, HybridCXR is demonstrated to be more computationally efficient than multi-model ensemble approaches, as all three branches are trained together in a single, end-to-end structure, avoiding the need for complex post-processing to merge outputs from independent models. The combination of multi-architecture design and the CWFL loss function thus leads not only to a numerical improvement in metrics but also to a more stable, adaptive, and generalizable solution for multi-label CXR classification.

5. Conclusion and Future Work

Multi-label classification of chest X-ray (CXR) images represents a formidable challenge in medical computer vision due to the subtle, overlapping signs of

pathologies and the severe inter-class imbalance in large-scale datasets, a key issue that traditionally causes standard deep learning models to fail in the reliable detection of rare diseases.

This study successfully addressed these challenges by proposing the HybridCXR model, a novel three-branch hybrid architecture. This model strategically integrates three complementary deep feature extractors—VGG16-BN (Classic CNN), ConvNeXt-Base (Modern CNN), and ViT-Base (Transformer)—within a single, unified framework. Crucially, the model utilizes the proposed Class-Weighted Focal Loss (CWFL) function, which provides an adaptive mechanism to focus the learning gradient on difficult and minority samples while maintaining optimal training stability.

Experimental validation on the NIH ChestX-ray14 dataset demonstrated the superiority of the proposed framework; the HybridCXR model achieved a state-of-the-art mean AUC-ROC of 0.870, significantly outperforming benchmark models such as CONVFCMAE (0.852) and CheX-DS (0.838). This performance, including a notable AUC of 0.920 on the rare Hernia class, confirmed the effectiveness of the multi-scale architecture and the dual-level gradient tuning provided by CWFL. This superiority is rooted in the model's dual scientific contribution: the "Triple-Backbone Hybrid" structure enhances feature diversity, and the CWFL function successfully balances the learning gradient at both the class-level and the instance-level. Furthermore, the end-to-end training of the entire HybridCXR structure offers greater computational efficiency compared to complex ensemble models.

While the model achieved strong results, future work should explore further evaluation on external datasets (e.g., CheXpert, MIMIC-CXR), the integration of multimodal learning (image + text reports), optimization of efficiency using lighter backbones, and the development of a novel multi-scale visualization technique tailored to this triple-branch architecture. Specifically, we plan to integrate Grad-CAM and self-attention maps from the VGG16-BN, ConvNeXt-Base and ViT-Base branches into a unified visualization that simultaneously highlights local, hierarchical and global regions contributing to each prediction. Such an XAI strategy would provide clinicians with intuitive explanations of the model's reasoning and clarify the clinical impact of each feature scale.

In conclusion, the HybridCXR model, by intelligently integrating architectural diversity with multi-level gradient tuning via CWFL, establishes a powerful foundation for the next generation of robust and reliable AI-based systems for automated lung disease diagnosis.

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Conflict of interest

The authors declare that they have no conflict of interest.

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Data Availability

The NIH ChestX-ray14 dataset used in this study is available at: https://nihcc.app.box.com/v/ChestXra
NIHCC

CRediT Author Statement

- Rouhalah Ebrahimi: Conceptualization, Methodology, Software, Validation, Writing - Original Draft
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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to assist with grammar checking and language polishing. After using this tool, the authors reviewed and edited all content thoroughly and take full responsibility for the content of the published article.

Appendix A. Example Appendix Section

Appendix text.

References

[1] Sharma A, Vans E, Shigemizu D, et al. Deep learning for automated pneumonia detection in chest radiographs: A review. *Comput Biol Med*. 2021;131:104258.

[2] Beam AL, Kohane IS. Big data and machine learning in health care. *JAMA*. 2018;319(13):1317-1318.

[3] Shamshad F, Khan S, Zamir SW, et al. Transformers in medical imaging: A survey. *Med Image Anal*. 2023;88:102802.

[4] Rajpurkar P, Irvin J, Zhu K, et al. CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv [Preprint]*. 2017. arXiv:1711.05225.

[5] Wang X, Peng Y, Lu L, et al. ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2017. p. 2097-2106.

[6] Shen D, Wu G, Suk HI. Deep learning in medical image analysis. *Annu Rev Biomed Eng*. 2017;19:221-248.

[7] Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv [Preprint]*. 2020. arXiv:2010.11929.

[8] Zhang Z, Liu Q, Wang Y. Medical image classification using synergic deep learning. *Med Image Anal*. 2019;54:10-19.

[9] Xiong K, Tu Y, Rao X, Zou X, Du Y. Multi-label disease detection in chest X-ray imaging using a fine-tuned ConvNeXtV2 with a customized classifier. *Informatics*. 2025;12(3):80.

[10] Li X, Liu Y, Xu X, Zhao X. CheX-DS: Improving chest X-ray image classification with ensemble learning based on DenseNet and Swin Transformer. *arXiv [Preprint]*. 2025. arXiv:2505.11168.

[11] Taslimi R, Sadeghi M. Multi-label classification on chest X-ray images with transformers (SwinCheX). *arXiv [Preprint]*. 2022. arXiv:2206.04246.

[12] Wu X, Li J, Zhang Y, Yuan L. CTransCNN: Combining transformer and CNN in multilabel medical image classification. *Knowl Based Syst*. 2023;276:110757.

[13] Ashraf SMN, Rahman MM, Rahman MM. SynthEnsemble: A fusion of CNN, vision transformer, and hybrid models for multi-label chest X-ray classification. *arXiv [Preprint]*. 2023. arXiv:2311.07750.

[14] Irvin J, Rajpurkar P, Ko M, et al. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. 2019;33(01):590-597.

[15] Guendel S, Grbic S, Georgescu B, et al. Learning to recognize abnormalities in chest radiographs with location-aware dense networks. In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*. 2018. p. 757-765.

[16] Yang S, Wu Y. Intelligent diagnosis model for chest X-ray image diseases based on convolutional neural network. *BMC Med Imaging*. 2025;25(1):263.

[17] Manzari ON, Ahmadabadi H, Kashiani H, et al. MedViT: A robust vision transformer for generalized medical image classification. *Comput Biol Med*. 2023;157:106791.

[18] Chen J, Lu Y, Yu Q, et al. TransUNet: Transformers make strong encoders for medical image segmentation. *arXiv [Preprint]*. 2021. arXiv:2102.04306.

[19] Pham HH, Le-Tien T, Nguyen HQ, et al. Interpretable thorax disease classification using attention-based CNNs. *IEEE Access*. 2023;11:11552-11563.

[20] Tang Y, Yang D, Li W, et al. MedViT: A hybrid vision transformer for medical image classification. *arXiv [Preprint]*. 2022. arXiv:2203.15012.

[21] Yao L, Prosky J, Lin A, et al. Learning long-range dependencies in chest X-ray classification using attention modules. *arXiv [Preprint]*. 2018. arXiv:1805.07335.

[22] Asadi KS, Gerard IJ, Kersten-Oertel M. Clinically-inspired hierarchical multi-label classification of chest X-rays with a penalty-based loss function (CIHMLC). *arXiv [Preprint]*. 2025. arXiv:2502.03591. doi:10.48550/arXiv.2502.03591.

[23] Bhusal D, Panday SP. Multi-label classification of thoracic diseases using dense convolutional network on chest radiographs. *arXiv [Preprint]*. 2022. arXiv:2202.03583. doi:10.48550/arXiv.2202.03583.

[24] Lin TY, Goyal P, Girshick R, He K, Dollár P. Focal loss for dense object detection. In: *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*. 2017. p. 2980-2988. doi:10.1109/ICCV.2017.324.

[25] Wang Y, Zhang X, Martel AL. Boosted focal loss for imbalanced medical image classification. *Med Image Anal*. 2022;80:102515.

[26] Chen B, Huang X, Li J, et al. Multi-scale CNN-Transformer networks for multi-label thorax disease classification. *IEEE J Biomed Health Inform*. 2023;27(9):4343-4354.

[27] Gao F, Sun J, Kang J, et al. TransMed: Transformer-based multimodal medical image fusion. *IEEE Trans Instrum Meas*. 2021;70:1-13.

[28] Haghighi F, Swarup S, Tiu E, et al. CheXbert: Combining BERT and CNNs for improved chest X-ray disease labeler. In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2021.

[29] Cao K, Wei C, Gaidon A, Arechiga N, Ma T. Learning imbalanced datasets with label-distribution-aware margin loss. In: *Advances in Neural Information Processing Systems*. 2019;32:1567-1578.

[30] Cui J, Li X, Wei Y, et al. Class-balanced focal loss for deep long-tailed recognition. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021. p. 14446-14455.

[31] Cui Y, Jia M, Lin TY, Song Y, Belongie S. Class-balanced loss based on effective number of samples. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019. p. 9268-9277. doi:10.1109/CVPR.2019.00949.

[32] Hanif MS, Bilal M, Alsaggaf AH, Al-Saggaf UM. Enhancing multi-label chest X-ray classification using an improved ranking loss. *Bioengineering (Basel)*. 2025;12(6):593. doi:10.3390/bioengineering12060593.

[33] Ridnik T, Ben-Baruch E, Noy A, Zelnik-Manor L. Asymmetric loss for multi-label classification. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2021. p. 82-91.